DATABASE TECHNOLOGIES LAB SHEET — 1

Course Code: MCA105P

Section A — Easy (20 Problems)

Q1. Write a DDL statement to create an empty table named departments with an integer dept_id as the primary key and a dept_name of up to 50 characters.

Syntax / Answer:

```
CREATE TABLE departments (dept_id INT PRIMARY KEY, dept_name VARCHAR(50));
```

Q2. Write a DDL query to create a table employees with an emp_id (Primary Key), first_name (VARCHAR 30), and salary (DECIMAL 10,2).

Syntax / Answer:

```
CREATE TABLE employees (emp_id INT PRIMARY KEY, first_name VARCHAR(30), salary DECIMAL(10,2));
```

Q3. Create a table projects with project_id (Primary Key), project_name (VARCHAR 100), and start_date using the standard DATE data type.

Syntax / Answer:

```
CREATE TABLE projects (project_id INT PRIMARY KEY, project_name VARCHAR(100), start_date DATE);
```

Q4. Create a table audit_logs storing log_id (Primary Key), action (TEXT), and created_at using TIMESTAMP.

Syntax / Answer:

```
CREATE TABLE audit_logs (log_id INT PRIMARY KEY, action TEXT, created_at TIMESTAMP);
```

Q5. Write a DDL query to create a table products with product_id (Primary Key), product_name (VARCHAR 50), and is_available using BOOLEAN.

Syntax / Answer:

```
CREATE TABLE products (product_id INT PRIMARY KEY, product_name VARCHAR(50), is_available
BOOLEAN);
```

Q6. Create a table system_settings with setting_id (SMALLINT Primary Key), setting_key (VARCHAR 50), and setting_value (VARCHAR 100).

Syntax / Answer:

```
CREATE TABLE system_settings (setting_id SMALLINT PRIMARY KEY, setting_key VARCHAR(50),
setting_value VARCHAR(100));
```

Q7. Write an ALTER command to add a new column email (VARCHAR 100) to the existing employees table.

Syntax / Answer:

```
ALTER TABLE employees ADD COLUMN email VARCHAR(100);
```

Q8. Write an ALTER statement to remove the is_available column from the products table.

Syntax / Answer:

```
ALTER TABLE products DROP COLUMN is_available;
```

Q9. Write the SQL DDL command to rename the table departments to dept_units.

Syntax / Answer:

```
RENAME TABLE departments TO dept_units;
```

Q10. Write a command to empty all records from audit_logs without dropping the table structure.

Syntax / Answer:

```
TRUNCATE TABLE audit_logs;
```

Q11. Create a table user_profiles with user_id (Primary Key), bio (TEXT), and profile_pic (BLOB).

Syntax / Answer:

```
CREATE TABLE user_profiles (user_id INT PRIMARY KEY, bio TEXT, profile_pic BLOB);
```

Q12. Create a table ticket_orders with ticket_id (Primary Key) and priority_level restricted to ENUM('Low', 'Medium', 'High').

Syntax / Answer:

```
CREATE TABLE ticket_orders (ticket_id INT PRIMARY KEY, priority_level
ENUM('Low','Medium','High'));
```

Q13. Write the DDL statement to drop the entire products table permanently from the database.

Syntax / Answer:

```
DROP TABLE products;
```

Q14. Write a command to change the data type of email in employees from VARCHAR(100) to VARCHAR(150).

Syntax / Answer:

```
ALTER TABLE employees MODIFY COLUMN email VARCHAR(150);
```

Q15. Create a table sensor_data with read_id (BIGINT Primary Key) and temperature (FLOAT).

Syntax / Answer:

```
CREATE TABLE sensor_data (read_id BIGINT PRIMARY KEY, temperature FLOAT);
```

Q16. Add a composite primary key to an existing table student_courses using columns student_id (INT) and course_id (INT).

Syntax / Answer:

```
ALTER TABLE student_courses ADD PRIMARY KEY (student_id, course_id);
```

Q17. Rename column first_name to given_name in the employees table using DDL.

Syntax / Answer:

```
ALTER TABLE employees RENAME COLUMN first_name TO given_name;
```

Q18. Write an ALTER command to drop the primary key constraint from the system_settings table.

Syntax / Answer:

```
ALTER TABLE system_settings DROP PRIMARY KEY;
```

Q19. Create a table clients with client_id (INT Primary Key) and registration_time (TIME).

Syntax / Answer:

```
CREATE TABLE clients (client_id INT PRIMARY KEY, registration_time TIME);
```

Q20. Re-add a primary key to column client_id in clients after table creation using an ALTER command.

Syntax / Answer:

```
ALTER TABLE clients ADD PRIMARY KEY (client_id);
```

Section B — Medium (20 Problems)

Q21. Create a table order_items representing a line-item in an order. Use a composite primary key consisting of order_id (INT) and item_id (INT), along with quantity (INT) and unit_price (DECIMAL 8,2).

Syntax / Answer:

```
CREATE TABLE order_items (order_id INT, item_id INT, quantity INT, unit_price DECIMAL(8,2),
PRIMARY KEY (order_id, item_id));
```

Q22. Create a table event_schedules with event_id (INT), venue_id (INT), event_date (DATE), and duration_hours (TINYINT). Make event_id and venue_id the composite primary key.

Syntax / Answer:

```
CREATE TABLE event_schedules (event_id INT, venue_id INT, event_date DATE, duration_hours TINYINT,
PRIMARY KEY (event_id, venue_id));
```

Q23. Modify the employees table to add two new columns in a single DDL statement: hire_date (DATE) and is_active (BOOLEAN).

Syntax / Answer:

```
ALTER TABLE employees ADD COLUMN hire_date DATE, ADD COLUMN is_active BOOLEAN;
```

Q24. In the table employees, drop column email and change salary data type to DECIMAL(12,2) in a single ALTER statement.

Syntax / Answer:

```
ALTER TABLE employees DROP COLUMN email, MODIFY COLUMN salary DECIMAL(12,2);
```

Q25. Given table inventory with columns warehouse_id (INT) and sku (VARCHAR 20), write an ALTER statement to add a composite primary key to both columns.

Syntax / Answer:

```
ALTER TABLE inventory ADD PRIMARY KEY (warehouse_id, sku);
```

Q26. Write a DDL command to change the bio column in user_profiles from TEXT to MEDIUMTEXT.

Syntax / Answer:

```
ALTER TABLE user_profiles MODIFY COLUMN bio MEDIUMTEXT;
```

Q27. Add a column middle_name (VARCHAR 30) to employees specifically positioned right after given_name.

Syntax / Answer:

```
ALTER TABLE employees ADD COLUMN middle_name VARCHAR(30) AFTER given_name;
```

Q28. Create a table geolocations with location_id (INT Primary Key), latitude (DECIMAL 9,6), and longitude (DECIMAL 9,6).

Syntax / Answer:

```
CREATE TABLE geolocations (location_id INT PRIMARY KEY, latitude DECIMAL(9,6), longitude
DECIMAL(9,6));
```

Q29. Create a table user_accounts with user_id (BIGINT Primary Key), account_type (ENUM('Standard', 'Premium', 'Admin')), and created_timestamp (TIMESTAMP).

Syntax / Answer:

```
CREATE TABLE user_accounts (user_id BIGINT PRIMARY KEY, account_type
ENUM('Standard','Premium','Admin'), created_timestamp TIMESTAMP);
```

Q30. Create a table device_logs with log_id (BIGINT Primary Key), ip_address (VARCHAR 45), and payload (LONGBLOB).

Syntax / Answer:

```
CREATE TABLE device_logs (log_id BIGINT PRIMARY KEY, ip_address VARCHAR(45), payload LONGBLOB);
```

Q31. Drop the existing primary key of order_items, add a new column order_item_id (INT) at the first position, and set it as the new primary key in a single ALTER command chain.

Syntax / Answer:

```
ALTER TABLE order_items DROP PRIMARY KEY, ADD COLUMN order_item_id INT FIRST, ADD PRIMARY KEY
(order_item_id);
```

Q32. Convert start_date column in table projects to project_year with data type YEAR using a single DDL statement.

Syntax / Answer:

```
ALTER TABLE projects CHANGE COLUMN start_date project_year YEAR;
```

Q33. Create a table document_store with doc_id (INT Primary Key), doc_title (VARCHAR 255), and content (LONGTEXT).

Syntax / Answer:

```
CREATE TABLE document_store (doc_id INT PRIMARY KEY, doc_title VARCHAR(255), content LONGTEXT);
```

Q34. Change column content in document_store to doc_body while maintaining its LONGTEXT type.

Syntax / Answer:

```
ALTER TABLE document_store RENAME COLUMN content TO doc_body;
```

Q35. Write a DDL command to create an empty table structure employees_archive identical in layout and constraints to the employees table.

Syntax / Answer:

```
CREATE TABLE employees_archive LIKE employees;
```

Q36. Write a DDL query to drop table employees_archive safely only if it exists in the schema.

Syntax / Answer:

```
DROP TABLE IF EXISTS employees_archive;
```

Q37. Create a table financial_ledger with entry_id (BIGINT Primary Key), debit (DECIMAL 15,4), and credit (DECIMAL 15,4).

Syntax / Answer:

```
CREATE TABLE financial_ledger (entry_id BIGINT PRIMARY KEY, debit DECIMAL(15,4), credit
DECIMAL(15,4));
```

Q38. Add transaction_date (DATETIME) as the absolute first column in the financial_ledger table using ALTER.

Syntax / Answer:

```
ALTER TABLE financial_ledger ADD COLUMN transaction_date DATETIME FIRST;
```

Q39. Create a table measurements with sample_id (INT Primary Key) and value (DOUBLE).

Syntax / Answer:

```
CREATE TABLE measurements (sample_id INT PRIMARY KEY, value DOUBLE);
```

Q40. Create a table app_users with user_id (INT Primary Key) and status_code (TINYINT).

Syntax / Answer:

```
CREATE TABLE app_users (user_id INT PRIMARY KEY, status_code TINYINT);
```

Section C — Hard (10 Problems)

Q41. Design a transactions table with an xact_id (BIGINT Primary Key), user_id (INT), amount (DECIMAL 10,2), channel (ENUM('Web', 'Mobile', 'ATM')), status (VARCHAR 20), and xact_time (TIMESTAMP).

Syntax / Answer:

```
CREATE TABLE transactions (xact_id BIGINT PRIMARY KEY, user_id INT, amount DECIMAL(10,2), channel
ENUM('Web','Mobile','ATM'), status VARCHAR(20), xact_time TIMESTAMP);
```

Q42. Create a high-throughput logging table system_events using event_id (BIGINT), node_id (SMALLINT), event_payload (TEXT), and logged_time (TIMESTAMP). Define a composite primary key on (event_id, node_id).

Syntax / Answer:

```
CREATE TABLE system_events (event_id BIGINT, node_id SMALLINT, event_payload TEXT, logged_time
TIMESTAMP, PRIMARY KEY (event_id, node_id));
```

Q43. Redesign existing table clients in a single ALTER statement: drop its current primary key, add region_code (CHAR 3), and set (client_id, region_code) as the new composite primary key.

Syntax / Answer:

```
ALTER TABLE clients DROP PRIMARY KEY, ADD COLUMN region_code CHAR(3), ADD PRIMARY KEY (client_id,
region_code);
```

Q44. You have a table raw_data with column temp_val (VARCHAR 50). In a single ALTER command, rename column temp_val to converted_val, change its type to DECIMAL(6,3), and add column processed_at (TIMESTAMP) right after it.

Syntax / Answer:

```
ALTER TABLE raw_data CHANGE COLUMN temp_val converted_val DECIMAL(6,3), ADD COLUMN processed_at TIMESTAMP AFTER converted_val;
```

Q45. Create a student_enrollments junction table with student_id (INT), course_id (INT), semester (VARCHAR 10), and enrollment_date (DATE). Set a 3-column composite primary key across (student_id, course_id, semester).

Syntax / Answer:

```
CREATE TABLE student_enrollments (student_id INT, course_id INT, semester VARCHAR(10), enrollment_date DATE, PRIMARY KEY (student_id, course_id, semester));
```

Q46. Refactor student_enrollments by dropping the composite key (student_id, course_id, semester), adding enrollment_id (BIGINT) as the first column, and establishing it as the new primary key in a single statement.

Syntax / Answer:

```
ALTER TABLE student_enrollments DROP PRIMARY KEY, ADD COLUMN enrollment_id BIGINT FIRST, ADD PRIMARY KEY (enrollment_id);
```

Q47. Create table file_metadata with file_id (INT Primary Key), file_path (VARCHAR 500), file_size_bytes (BIGINT), and checksum (CHAR 64).

Syntax / Answer:

```
CREATE TABLE file_metadata (file_id INT PRIMARY KEY, file_path VARCHAR(500), file_size_bytes BIGINT, checksum CHAR(64));
```

Q48. Modify file_metadata schema in one ALTER statement: Change file_id from INT to BIGINT, alter file_path to TEXT, and drop column checksum.

Syntax / Answer:

```
ALTER TABLE file_metadata MODIFY COLUMN file_id BIGINT, MODIFY COLUMN file_path TEXT, DROP COLUMN checksum;
```

Q49. Write the SQL DDL statements to: (1) Create a table sensor_readings with composite PK (sensor_id INT, recorded_at TIMESTAMP) and column reading (FLOAT), and (2) Rename sensor_readings to historical_sensor_readings.

Syntax / Answer:

```
CREATE TABLE sensor_readings (sensor_id INT, recorded_at TIMESTAMP, reading FLOAT, PRIMARY KEY (sensor_id, recorded_at));
RENAME TABLE sensor_readings TO historical_sensor_readings;
```

Q50. Write the SQL DDL statements to: (1) Create table application_logs with log_id (BIGINT Primary Key), level (ENUM('DEBUG', 'INFO', 'WARN', 'ERROR')), message (TEXT), execution_time (DOUBLE), and created_at (DATETIME), and (2) Modify the level column to VARCHAR(10).

Syntax / Answer:

```
CREATE TABLE application_logs (log_id BIGINT PRIMARY KEY, level ENUM('DEBUG','INFO','WARN','ERROR'), message TEXT, execution_time DOUBLE, created_at DATETIME);
ALTER TABLE application_logs MODIFY COLUMN level VARCHAR(10);
```